

CUET PG COMPUTER SCIENCE (SCQP09)

Data Structures & Algorithms – Arrays Practice Sheet (Easy Level)

Topic: Arrays & Memory

Total Questions: 30 MCQs

Difficulty: Easy / Basics

Target Score: 70%+ (21+ Marks)

1. Which of the following best describes an Array data structure?

- (A) Non-linear collection of elements
- (B) Homogeneous collection stored in contiguous memory
- (C) Heterogeneous collection stored dynamically
- (D) Hierarchical organization of nodes

2. What is the time complexity to access an element in an array given its index?

- (A) $O(1)$
- (B) $O(n)$
- (C) $O(\log n)$
- (D) $O(n^2)$

3. In standard zero-indexed arrays, what is the index of the 5th element?

- (A) 5
- (B) 4
- (C) 6
- (D) 0

4. An array `arr` has 10 elements. What are the valid index bounds in C/C++/Java?

- (A) 1 to 10
- (B) 0 to 10
- (C) 0 to 9
- (D) -1 to 9

5. What is the worst-case time complexity of searching an element in an unsorted array using Linear Search?

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n \log n)$

6. Given a 1D array starting at base address 2000, where each element takes 4 bytes. What is the address of element at index 3?

- (A) 2004
- (B) 2008
- (C) 2012
- (D) 2016

7. Insertion of an element at the beginning of an array of size n takes:

- (A) $O(1)$ time
- (B) $O(n)$ time due to shifting
- (C) $O(\log n)$ time
- (D) $O(n^2)$ time

8. Insertion of an element at the end of an array (without overflow) takes:

- (A) $O(1)$ time
- (B) $O(n)$ time
- (C) $O(\log n)$ time
- (D) $O(n^2)$ time

9. Which of the following operations takes $O(1)$ time in a standard array?

- (A) Deleting the first element
- (B) Updating element at index i
- (C) Searching for a target value
- (D) Inserting at index i

10. How is a 2D array declared with 3 rows and 4 columns in C?

- (A) `int arr[3, 4];`
- (B) `int arr[3][4];`
- (C) `int arr(3)(4);`
- (D) `array arr[3][4];`

11. In Row-Major Order, how are 2D array elements stored in memory?

- (A) Column by column
- (B) Row by row sequentially
- (C) Diagonal by diagonal
- (D) Randomly in RAM blocks

12. What is the total number of elements in a 2D array defined as `int A[4][5]`?

- (A) 9
- (B) 20
- (C) 16
- (D) 25

13. Binary Search can be applied on an array ONLY if the array is:

- (A) Unsorted
- (B) Sorted
- (C) Containing only positive numbers
- (D) Of even size

14. What is the time complexity of Binary Search on a sorted array of size n ?

- (A) $O(1)$
- (B) $O(n)$
- (C) $O(\log n)$
- (D) $O(n \log n)$

15. What will be the output of the following code snippet?

```
int arr[5] = {10, 20, 30}; printf("%d", arr[3]);
```

- (A) Garbage value / 0
- (B) 30
- (C) 10
- (D) Compilation Error

16. What is the main drawback of a static array compared to a linked list?

- (A) Slower random access
- (B) Fixed size defined at compile/allocation time
- (C) Higher memory overhead per element
- (D) Elements cannot be traversed sequentially

17. Which formula calculates the number of elements in an array with lower bound LB and upper bound UB?

- (A) $UB - LB$
- (B) $UB - LB + 1$
- (C) $UB + LB$
- (D) $UB - LB - 1$

18. Traversing an array means:

- (A) Sorting all elements in ascending order
- (B) Accessing each element exactly once
- (C) Inserting new elements into empty positions
- (D) Deleting redundant values

19. If an array `arr = [4, 1, 8, 3]` is given, what is the value of `arr[1] + arr[3]`?

- (A) 12
- (B) 4
- (C) 9
- (D) 5

20. In C language, the array name acts as a constant pointer to its:

- (A) Last element
- (B) First element (Base Address)
- (C) Middle element
- (D) Size of the array

21. What is the spatial auxiliary complexity required to find the Maximum element in a 1D array?

- (A) $O(1)$
- (B) $O(n)$
- (C) $O(\log n)$
- (D) $O(n^2)$

22. What happens if you try to access an index out of array bounds in C programming?

- (A) Automatically throws `ArrayOutOfBoundsException`
- (B) Undefined Behavior / Undefined Garbage Output
- (C) Resizes the array automatically
- (D) Returns `NULL` always

23. Reversing an array of size n in-place takes how much minimum time?

- (A) $O(1)$
- (B) $O(n)$
- (C) $O(n \log n)$
- (D) $O(n^2)$

24. What is the lower bound address calculation formula for Row-Major 2D array element $A[i][j]$ (0-based indexing)?

- (A) $\text{Base} + (i * \text{cols} + j) * \text{size}$
- (B) $\text{Base} + (j * \text{rows} + i) * \text{size}$
- (C) $\text{Base} + (i + j) * \text{size}$
- (D) $\text{Base} + (i * j) * \text{size}$

25. Which sorting algorithm repeatedly swaps adjacent elements if they are in wrong order?

- (A) Selection Sort
- (B) Bubble Sort
- (C) Insertion Sort
- (D) Quick Sort

26. An array $A = [2, 4, 6, 8, 10]$. How many comparisons are made by Linear Search to search for key = 8?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

27. Which of the following data structure uses contiguous array as its primary underlying structure?

- (A) ArrayList / Vector
- (B) Doubly Linked List
- (C) Binary Search Tree
- (D) Graph Adjacency List

28. What is the maximum number of comparisons in Binary Search for an array of 16 sorted elements?

- (A) 4
- (B) 5
- (C) 16
- (D) 8

29. In Column-Major Order storage of a 2D array, elements are stored:

- (A) Column by column in contiguous memory
- (B) Row by row
- (C) Anti-diagonally
- (D) In non-contiguous heap blocks

30. What is the time complexity of finding the Minimum element in a sorted array (ascending order)?

- (A) $O(1)$
- (B) $O(n)$
- (C) $O(\log n)$
- (D) $O(n \log n)$

Answer Key & Brief Explanations (Questions 1 – 30)

Q1. (B)

Arrays store homogeneous elements in contiguous memory locations.

Q2. (A)

Direct indexing allows $O(1)$ random access via base address formula.

Q3. (B)

Index = position - 1. So 5th element is at index 4.

Q4. (C)

Indices range from 0 to $n-1$ (0 to 9 for size 10).

Q5. (C)

Worst case checks all n elements, taking $O(n)$ time.

Q6. (C)

Address = $2000 + (3 * 4) = 2000 + 12 = 2012$.

Q7. (B)

Inserting at front requires shifting all existing n elements right.

Q8. (A)

Inserting at the tail requires no element shifting, taking $O(1)$.

Q9. (B)

Updating element using `arr[i] = val` takes constant $O(1)$ time.

Q10. (B)

C uses bracket syntax `[rows][cols]` for 2D arrays.

Q11. (B)

Row-Major Order stores elements row after row in memory.

Q12. (B)

Total elements = Rows * Columns = $4 * 5 = 20$.

Q13. (B)

Binary search requires a sorted search space to eliminate halves.

Q14. (C)

Dividing search space in half repeatedly gives $O(\log n)$ complexity.

Q15. (A)

Uninitialized array slots in C default to 0 (or garbage depending on scope).

Q16. (B)

Static arrays cannot grow or shrink dynamically at runtime.

Q17. (B)

Length = Upper Bound - Lower Bound + 1.

Q18. (B)

Traversal refers to visiting every element in array once.

Q19. (B)

`arr[1] = 1`, `arr[3] = 3`. Sum = $1 + 3 = 4$.

Q20. (B)

Array name decays to pointer pointing to base address (`&arr[0]`).

Q21. (A)

Only 1 variable needed to track max, requiring $O(1)$ space.

Q22. (B)

C lacks bound checking, causing Undefined Behavior/garbage.

Q23. (B)

Two-pointer swap takes $n/2$ operations, which is $O(n)$ time.

Q24. (A)

Row-Major formula: $\text{Base} + (i * \text{cols} + j) * \text{element_size}$.

Q25. (B)

Bubble Sort repeatedly compares and swaps adjacent elements.

Q26. (C)

Key 8 is at index 3 (4th element), requiring 4 comparisons.

Q27. (A)

Dynamic arrays/ArrayLists use contiguous arrays under the hood.

Q28. (B)

$\log_2(16) + 1 = 4 + 1 = 5$ comparisons in worst case.

Q29. (A)

Column-Major stores complete columns sequentially in memory.

Q30. (A)

First element `arr[0]` is minimum in ascending array $\rightarrow O(1)$.

